1 Generated reference tables

The tables in this appendix are generated from the source tree by `scripts/gen_report_tables.py`. They are not maintained by hand: a renamed flag, a new cycle-slip detector, or a changed state-key delimiter appears here on the next run of the generator. The fragments under `generated/` are `.gitignore`d and rebuilt on demand (`build.ps1 -Regen` / `build.sh --regen`); run the generator with `--check` to verify they match the current source tree.

1.1 Processing modes

Table 1: `ProcessingMode` – the estimation algorithm selector.

Value	Meaning
`PPP`	Precise Point Positioning (single receiver, external products).
`PPP_RTK`	PPP with RTK enhancements (single receiver, uncombined).
`BaselineSingle`	Double-differenced single baseline: one rover + one reference station.
`BaselineMulti`	Double-differenced multi-receiver network (star topology: N rovers against one reference).
`PPP_RTK_Network`	Minimum-constraint S-system network PRODUCER: joint multi-station estimator emitting satellite clock + code/phase bias (+ atmosphere) under a declared S-basis.
`PPP_RTK_User`	PPP-RTK CONSUMER of network PPP_RTK_Network products (post-processing closed-loop counterpart): uncombined PPP-RTK + network SSR bias/atmosphere consume + S-basis cross-check.

1.2 Execution mode and data source

Table 2: `AppMode` – top-level post-processing vs. real-time dispatch.

Value	Meaning
`PostProcessing`	Batch processing from RINEX / product files.
`RealTime`	Streaming / live processing from RTCM data.

Table 3: `DataSource` – bundled fixtures vs. auto-download.

Value	Meaning
`Internal`	Use files under internal_data_root only.
`Production`	Auto-download from CDDIS into data_root.

1.3 Differencing regime

Table 4: `DifferencingMode` – the measurement differencing regime.

Value	Meaning
`None`	Raw observables (PPP / PPP-RTK without SD).
`BetweenSatellite`	Per-constellation single difference vs. pivot (PPP-RTK SD).
`DoubleDifference`	Between-receiver + between-satellite (baseline processing).

1.4 Config-to-differencing mapping

The differencing regime is resolved once per run by the factory below (verbatim from `MeasurementBuilder.cpp`):

```
DifferencingMode differencingModeFor(const GnssConfig& config) {
    // The network producer runs on undifferenced + uncombined rows and estimates
    // the satellite-side bias lumps explicitly -- it must NEVER apply receiver-side
    // BSSD even if a between_satellite_sd toggle is present in its config.
    if (config.getProcessingMode() == ProcessingMode::PPP_RTK_Network)
                                        return DifferencingMode::None;
    if (config.isBaselineMode())        return DifferencingMode::DoubleDifference;
    if (config.enableBetweenSatelliteSd()) return DifferencingMode::BetweenSatellite;
    return DifferencingMode::None;
}
```

1.5 Feature toggles

These are the loop- and state-vector-level switches.

Table 5: Top-level boolean feature toggles, parsed from the `jval<bool>` calls in `GnssConfigLoad.cpp` and `GnssConfigLoadNetwork.cpp`. The legacy alias is the pre-refactor flat key still accepted as a fallback.

JSON path	Default	Legacy alias
`estimation.accumulate_phase_windup`	off	`accumulate_phase_windup`
`estimation.adaptive_iono_process_noise.trace`	off	`adaptive_iono_q_trace`
`estimation.aiding.baro`	off	`aiding_baro`
`estimation.aiding.nhc`	off	`aiding_nhc`
`estimation.aiding.vision`	off	`aiding_vision`
`estimation.aiding.wheel`	off	`aiding_wheel`
`estimation.aiding.zupt`	off	`aiding_zupt`
`estimation.ambiguity_resolution.enable_partial_ar`	off	`ar_enable_partial_ar`
`estimation.ambiguity_resolution.wl_only`	off	`ar_wl_only`

JSON path	Default	Legacy alias
estimation.between_satellite_sd.enable_glonass	off	enable_glonass_bssd
estimation.between_satellite_sd.enabled	off	enable_between_satellite_sd
estimation.cycle_slip.ewl	off	cycle_slip_ewl
estimation.cycle_slip.gf	off	cycle_slip_gf
estimation.cycle_slip.graphic	off	cycle_slip_graphic
estimation.cycle_slip.mw	on	cycle_slip_mw
estimation.cycle_slip.turbo_edit	off	cycle_slip_turbo_edit
estimation.cycle_slip.turbo_edit_bridge_lli	off	cycle_slip_turbo_edit_bridge_lli
estimation.cycle_slip.turbo_edit_mw_only_veto	on	cycle_slip_turbo_edit_mw_only_veto
estimation.defer_until_corrections_ready.drop_broadcast_grade	off	defer_until_corrections_ready_drop_broadcast_grade
estimation.defer_until_corrections_ready.require_all_above_mask	off	defer_until_corrections_ready_require_all_above_mask
estimation.enable_cio_frame	off	enable_cio_frame
estimation.enable_data_snooping	off	enable_data_snooping
estimation.enable_full_r_kernel	off	enable_full_r_kernel
estimation.enable_huber_innovation	off	enable_huber_innovation
estimation.enable_huber_r_inflation	on	enable_huber_r_inflation
estimation.enable_if_slip_detector	off	enable_if_slip_detector
estimation.enable_iono_rate	off	enable_iono_rate
estimation.enable_subcycle_slip_suppression	on	enable_subcycle_slip_suppression
estimation.enable_variance_factor	off	enable_variance_factor
estimation.estimate_ifb	off	estimate_ifb
estimation.estimate_phase_clock	off	estimate_phase_clock
estimation.estimate_sat_biases	off	estimate_sat_biases
estimation.estimate_sat_clocks	off	estimate_sat_clocks
estimation.estimate_sat_phase_biases	off	estimate_sat_phase_biases
estimation.estimate_tropo_gradients	off	estimate_tropo_gradients
estimation.estimate_velocity	off	estimate_velocity
estimation.global_vtec_sh.sun_fixed	on	global_vtec_sh_sun_fixed
estimation.glonass_ifb.enable	off	enable_glonass_ifb
estimation.lock_signals_to_clock_ref	off	lock_signals_to_clock_ref
estimation.promote_sat_code_bias	off	promote_sat_code_bias
estimation.require_bia_for_signals	off	require_bia_for_signals
estimation.residual_history.enabled	off	residual_history_enabled
network.ambiguity_resolution.enable_partial_ar	on	network_ar_enable_partial_ar
network.ambiguity_resolution.wl_cascade	off	network_ar_wl_cascade
network.carry_conditional_coupling	off	carry_conditional_coupling

JSON path	Default	Legacy alias
network.carry_coupling_joint_whiten	off	carry_coupling_joint_whiten
network.carry_coupling_sh_marginalize	off	carry_coupling_sh_marginalize
network.carry_coupling_time_update	off	carry_coupling_time_update
network.carry_invariant	off	network_carry_invariant
network.carry_invariant_emit_rebase	on	network_carry_invariant_emit_rebase
network.carry_invariant_honesty	off	network_carry_invariant_honesty
network.carry_invariant_innov_calibrate	off	network_carry_invariant_innov_calibrate
network.carry_invariant_innov_gate	off	network_carry_invariant_innov_gate
network.carry_invariant_innov_refold	off	network_carry_invariant_innov_refold
network.carry_vce	off	carry_vce
network.carry_vce_apply	off	carry_vce_apply
network.carry_zwd	off	carry_zwd
network.clock_datum_broadcast_anchored	off	network_clock_datum_broadcast_anchored
network.clock_datum_sum_to_zero	off	network_clock_datum_sum_to_zero
network.clock_disc_on_set_change	off	network_clock_disc_on_set_change
network.clock_model	off	network_clock_model
network.clock_monitor	off	network_clock_monitor
network.clock_monitor_couple_raim	off	network_clock_monitor_couple_raim
network.clock_runaway_check	off	network_clock_runaway_check
network.clock_signal_basis_hold	off	network_clock_signal_basis_hold
network.code_bias_disc_jump_test	off	network_code_bias_disc_jump_test
network.code_datum_time_linked	off	network_code_datum_time_linked
network.cold_clock_seed	off	cold_clock_seed
network.consume_phase_bias_sigma	off	network_consume_phase_bias_sigma
network.consume_ura	off	network_consume_ura
network.cycle_slip.if_phase_td	off	network_cycle_slip_if_phase_td
network.cycle_slip.mw_detect	off	network_cycle_slip_mw_detect

JSON path	Default	Legacy alias
network.cycle_slip.turbo_edit	off	network_cycle_slip_turbo_edit
network.emit_clock_disc	off	network_emit_clock_disc
network.emit_clock_rate	off	network_emit_clock_rate
network.emit_combined_sigma	off	network_emit_combined_sigma
network.emit_epoch_status_record	off	network_emit_epoch_status_record
network.emit_generation_id	off	network_emit_generation_id
network.emit_generation_time	off	network_emit_generation_time
network.emit_pb_filter	off	network_emit_pb_filter
network.emit_pb_filter_emit_sigma	off	network_emit_pb_filter_emit_sigma
network.emit_pb_filter_gate_v2	off	network_emit_pb_filter_gate_v2
network.emit_phase_bias_sigma	off	network_emit_phase_bias_sigma
network.emit_ura	off	network_emit_ura
network.estimate_glonass	off	estimate_glonass
network.estimate_glonass	off	estimate_glonass
network.estimate_iono_sh	off	estimate_iono_sh
network.estimate_sat_code_bias	off	estimate_sat_code_bias
network.exclude_unmodeled_antenna_stations	off	network_exclude_unmodeled_antenna_stations
network.iono_constraint_deweight_adaptive	off	iono_constraint_deweight_adaptive
network.iono_supply_formal_sigma	off	iono_supply_formal_sigma
network.lane_carry.bridge_meas	on	network_lane_carry_bridge_meas
network.lane_carry.lane_meas	on	network_lane_carry_lane_meas
network.mw_wl.bridge_sample_debias	off	network_mw_wl_bridge_sample_debias
network.mw_wl.d5_dump	off	network_mw_wl_d5_dump
network.mw_wl.ewl_enable	off	network_mw_wl_ewl_enable
network.mw_wl.ewl_fold_rows	off	network_mw_wl_ewl_fold_rows
network.mw_wl.f6_dump	off	network_mw_wl_f6_dump
network.mw_wl.fix_enable	off	network_mw_wl_fix_enable
network.mw_wl.lane_trace	off	network_mw_wl_lane_trace
network.mw_wl.uwl_cold_fix	off	network_mw_wl_uwl_cold_fix
network.mw_wl.uwl_dual_top	off	network_mw_wl_uwl_dual_top
network.mw_wl.uwl_enable	off	network_mw_wl_uwl_enable
network.mw_wl.uwl_family_pairs	off	network_mw_wl_uwl_family_pairs

JSON path	Default	Legacy alias
network.mw_wl.uwl_mid_enable	off	network_mw_wl_uwl_mid_enable
network.mw_wl.uwl_sib_gate	off	network_mw_wl_uwl_sib_gate
network.mw_wl.wl_inject_enable	off	network_mw_wl_wl_inject_enable
network.parallel_carry	off	parallel_carry
network.parallel_reduce	off	parallel_reduce
network.parallel_seam	off	parallel_seam
network.parallel_stations	on	parallel_stations
network.per_const_withhold	off	network_per_const_withhold
network.phase_bias_exclude_unmatched_antenna	off	network_phase_bias_exclude_unmatched_antenna
network.phase_datum_ensemble	off	network_phase_datum_ensemble
network.phase_datum_eq34	off	network_phase_datum_eq34
network.phase_datum_eq34_carry_shift	off	network_phase_datum_eq34_carry_shift
network.phase_datum_eq34_extend_only	off	network_phase_datum_eq34_extend_only
network.phase_datum_eq34_longevity_election	off	network_phase_datum_eq34_longevity_election
network.phase_datum_eq34_pin_round	on	network_phase_datum_eq34_pin_round
network.phase_datum_prioritized	off	network_phase_datum_prioritized
network.phase_datum_sticky	on	network_phase_datum_sticky
network.phase_leveling	off	phase_leveling
network.phase_leveling_refloat_at_convergence	off	phase_leveling_refloat_at_convergence
network.phase_leveling_warmup_on_convergence	off	phase_leveling_warmup_on_convergence
network.pivot_vanish_return_signal	off	network_pivot_vanish_return_signal
network.product_flush_per_epoch	off	product_flush_per_epoch
network.raim	off	network_raim
network.raim_attribution	off	network_raim_attribution
network.raim_redundancy_normalize	off	network_raim_redundancy_normalize
network.raim_refloat_maturity_gate	off	network_raim_refloat_maturity_gate
network.raim_surgical_station	on	network_raim_surgical_station
network.raim_within_epoch	off	network_raim_within_epoch
network.require_signal_integer_for_ar	off	network_require_signal_integer_for_ar

JSON path	Default	Legacy alias
network.scoped_handover_response	off	network_scoped_handover_response
network.serialize_disc_counters	off	network_serialize_disc_counters
network.serialize_sbasis_contract	off	serialize_sbasis_contract
network.service_mode	off	network_service_mode
network.shared_carry_full_cov	off	shared_carry_full_cov
network.station_pd_pin_aware	off	network_station_pd_pin_aware
network.sticky_pivot	off	network_sticky_pivot
network.trace_anchor_info	off	trace_anchor_info
network.trace_clock_full	off	trace_clock_full
network.trace_dd_fractional	off	trace_dd_fractional
network.trace_epoch_status	off	trace_epoch_status
network.trace_estimability	off	trace_estimability
network.trace_events	off	trace_events
network.trace_flag_rates	off	trace_flag_rates
network.trace_ionosphere	off	trace_ionosphere
network.trace_section_timers	off	trace_section_timers
network.trace_slip_events	off	trace_slip_events
network.trace_slip_raw	off	trace_slip_raw
network.trace_ssr_consumer	off	trace_ssr_consumer
network.trace_station_residuals	off	trace_station_residuals
network.trace_stations	on	trace_stations
network.trace_zwd	off	trace_zwd
network.ura_retro_bump	off	network_ura_retro_bump
network.use_sif_coordinates	off	use_sif_coordinates
network.write_run_manifest	off	write_run_manifest
processing.ar_ignore_mw_wl	off	ar_ignore_mw_wl
processing.auto_report	off	auto_report
processing.enable_neq_export	off	enable_neq_export
processing.enable_noncausal_slip_bridge	off	enable_noncausal_slip_bridge
processing.enable_obs_skip_to_latest	off	enable_obs_skip_to_latest
processing.enable_obs_ssr_sync	off	enable_obs_ssr_sync
processing.enable_orbit_normal_attitude	off	enable_orbit_normal_attitude
processing.enable_ssr_bias_max_age	off	enable_ssr_bias_max_age
processing.enable_station_info	off	enable_station_info
processing.enable_trig_sp3_interp	off	enable_trig_sp3_interp
processing.exclude_eclipsing_satellites	off	exclude_eclipsing_satellites
processing.inhouse_bias_estimation	off	inhouse_bias_estimation
processing.network_consumer.bias	on	network_consumer_bias
processing.network_consumer.clock	on	network_consumer_clock
processing.network_consumer.enable	off	network_consumer_enable

JSON path	Default	Legacy alias
processing.network_consumer. undifferenced	on	network_consumer_ undifferenced
processing.relinearize_geometry	off	relinearize_geometry
processing.report_async	off	report_async
processing.run_smoother	off	run_smoother
processing.ssr_bias_band_fallback	off	ssr_bias_band_fallback
processing.uncombined_enabled	off	uncombined_enabled
receiver.enable_antenna_orientation	off	enable_antenna_orientation
receiver.enable_antenna_substitution	off	enable_antenna_ substitution
receiver.enable_itrf_velocity	off	enable_itrf_velocity

1.6 Model-correction switches

Table 6: Physical model on/off switches (ModelCorrections). These gate the correction terms applied to the undifferenced observation model.

Field	Default	Effect
enable_troposphere	on	Saastamoinen ZHD + GMF dry mapping (slant hydrostatic delay).
enable_klobuchar_ ionosphere	off	Klobuchar ionospheric delay model (single-frequency fallback only). IF combinations and DD short-baselines cancel ionosphere geometrically; PPP-RTK uncombined estimates per-satellite slant iono states. In all three of those configs the Klobuchar model is a no-op — but constructing the Ionosphere instance still pays for an extra Model::Geometry load and the Klobuchar coefficient parse. Default off. Enable only for single-frequency PPP runs that need a-priori iono.

Field	Default	Effect
enable_higher_order_ionosphere	off	Apply second-order (B·cos(θ)·STEC/f^3) and third-order (η·STEC^2/f^4) ionospheric delays as a per-signal observation-level correction, evaluated at the GIM-derived IPP using a centered tilted dipole geomagnetic field. Requires a loaded IONEX GIM as the absolute-STEC source; the engine auto-loads the GIM when this flag is on even if estimation.ionosphere_weighting.enabled is false. No-op when no GIM resolves (warning logged at startup). Magnitudes: I_2 ˜1–2 cm at L1 during high TEC, I_3 ˜mm; the IF combination only partially cancels these, so the largest visible benefit is in standard PPP (IF). For PPP-RTK uncombined and short-baseline DD the higher-order term mostly cancels in the SD / DD differencing, so this flag is closer to a consistency knob than a bias remover. Default off.
enable_relativistic_correction	on	Periodic relativistic clock correction (-2 rv / c).
enable_shapiro_delay	on	Shapiro (gravitational path) delay.
enable_solid_earth_tide	on	IERS 2010 solid Earth tide displacement.
enable_ocean_tide_loading	on	Ocean tidal loading displacement (BLQ).
enable_pole_tide	on	Pole tide displacement (from ERP): IERS 2010 Sec. 7.1.4 rotational deformation.
enable_ocean_pole_tide	off	Ocean pole tide load displacement (IERS 2010 Sec. 7.1.5 / Desai 2002), added on top of enable_pole_tide. Requires the opole load grid (files.opt_grid_path or ¡data¿/static/opole/). ˜0.1-2 mm; default off keeps runs byte-identical.
enable_atmospheric_loading	off	Atmospheric pressure loading (S1/S2 thermal-tide site displacement). Requires per-station S1/S2 load coefficients (files.atm_tide_grid_path or ¡data¿/static/apl/). Sun-synchronous → does NOT average out in static daily PPP. Default off keeps runs byte-identical.
enable_satellite_pco	on	Satellite antenna phase centre offset.
enable_satellite_pcv	on	Satellite antenna phase centre variation.
enable_receiver_pco	on	Receiver antenna phase centre offset.
enable_receiver_pcv	on	Receiver antenna phase centre variation.
enable_receiver_cross_system_atx_fallback	off	RECEIVER-ONLY: when the receiver ATX entry lacks a non-GPS frequency, substitute the nearest GPS band (E1-¿G01, E5a-¿G05, B1C-¿G01, B2a-¿G05, ...). Valid: receiver phase pattern is frequency-dependent on shared hardware. NEVER applied to the satellite antenna (E01 and G01 are different spacecraft). Opt-in.

Field	Default	Effect
enable_atx_frequency_fallback	on	When the ATX entry doesn't have PCO/PCV for the requested frequency (common for L5 / E5* / B2a on older satellites or receiver antennas), look up the nearest available frequency on the same antenna and use its values instead of silently skipping the correction. Cross-frequency PCO drift on a single physical antenna is typically ¡1 cm — better than the few-cm vertical residual that comes from skipping a frequency entirely. Set false to revert to the legacy "skip if missing" behavior for A/B comparison.
require_atx_coverage	off	When true, drop observations whose (sat, signal) lacks complete ATX coverage on **both** the satellite and receiver entries — i.e., satellite PCO+PCV and receiver PCO+PCV must all resolve to non-empty entries for the row's exact frequency, *without* enable_atx_frequency_fallback being used to substitute a neighbouring frequency. Default false (legacy behaviour: observations are kept and any missing PCO/PCV is silently zeroed). Set true when you've explicitly chosen 'signal_mode: "all"' and want the ATX file to be the authoritative whitelist, dropping rows that would otherwise process without proper antenna calibration.
require_atx_coverage_for_ar	off	When true, AR (WL constraints, LAMBDA NL search) skips (sat, signal) pairs lacking strict ATX coverage even though the underlying observations are still used in the float Kalman update. Softer counterpart to require_atx_coverage: lets you keep "all"-signal float estimation alive while restricting integer fixing to the ATX-curated subset. Default false (no AR-side gating).
restrict_ar_for_substituted_atx	off	When true, AR additionally skips (sat, signal) pairs whose ATX correction was computed via the frequency-fallback substitution chain (rather than an exact entry). Targets specifically the cases where the antenna model is approximate; signals lacking ATX entirely (no substitution applied) are still allowed. Use alongside or instead of require_atx_coverage_for_ar depending on which kind of imperfection you want to gate on. Default false.
loosen_amb_q_for_substituted_atx	off	When true, ambiguity states whose underlying (sat, signal) had their ATX correction frequency-fallback-substituted use a looser process noise (ambiguity_q_substituted_atx) instead of the standard process_noise.ambiguity. Compensates for the small but non-zero modeling error introduced by the substituted PCV/PCO. Default false.

Field	Default	Effect
apply_phase_shifts	off	When true, RINEX phase observations have the receiver-applied "SYS / PHASE SHIFT" values subtracted at parse time, recovering the unshifted phase convention. Default false because most modern BIA products (CODE, CAS, GFZ) are calibrated against the receiver-shifted phases as recorded in RINEX, so leaving the shifts in place is the standard PPP convention. Set true only when your bias product is explicitly calibrated for the unshifted convention, or when running without external BIA and using in-house bias estimation that benefits from a uniform shift-free input.
enable_phase_windup	on	Carrier-phase wind-up correction.
enable_code_phase_bias	on	Code/phase bias corrections from BIA product files.
bias_coverage_requires_apply	off	When true, the "already corrected?" gates in
refine_emission_time	on	When true, Geometry::computeSatelliteGeometry runs a

1.7 Logging and trace channels

Table 7: Logging and CSV-trace channels (LoggingConfig). Each toggles one spdlog file logger or one per-epoch CSV trace.

Field	Default	Channel
enable_main_log	on	logs/main_log.log
enable_gnss_config	on	logs/gnss_config.log
enable_kalman_filter	on	logs/kalman_filter.log
enable_smoother	on	logs/smoother.log
enable_state_manager	on	logs/state_manager.log
enable_ppp_model	on	logs/ppp_model.log
enable_measurement_model	on	logs/measurement_model.log
enable_mw_slip_detector	on	logs/mw_slip_detector.log
enable_if_slip_detector	on	logs/if_slip_detector.log
enable_gf_slip_detector	on	logs/gf_slip_detector.log
enable_turbo_edit_resolver	on	logs/turbo_edit_resolver.log
enable_cycle_slip_manager	on	logs/cycle_slip_manager.log
enable_ambiguity_resolver	on	logs/ambiguity_resolver.log
enable_corrections	on	logs/corrections.log

Field	Default	Channel
enable_geometry	on	logs/geometry.log
enable_troposphere	on	logs/troposphere.log
enable_ionosphere	on	logs/ionosphere.log
enable_antenna	on	logs/antenna.log
enable_antenna_model	on	logs/antenna_model.log
enable_ocean_tide	on	logs/ocean_tide.log
enable_planetary_model	on	logs/planetary_model.log
enable_stochastic_model	on	logs/stochastic_model.log
enable_rinex_parser	on	logs/rinex_parser.log
enable_product_parser	on	logs/product_parser.log
enable_product_cache	on	logs/product_cache.log
enable_resolver	on	logs/resolver.log
enable_state_initializer	on	logs/state_initializer.log
enable_bssd_pivot_manager	on	logs/bssd_pivot_manager.log
enable_bssd_transform	on	logs/bssd_transform.log
enable_ppp_utils	on	logs/ppp_utils.log
enable_downloader	on	logs/downloader.log
enable_ntrip	on	logs/ntrip.log
enable_rt_engine	on	logs/rt_engine.log
enable_imu	on	logs/imu.log
enable_state_trace	on	logs/state_trace.csv
enable_ambiguity_trace	on	logs/ambiguity_trace.csv
enable_ionosphere_trace	on	logs/ionosphere_trace.csv
enable_sat_bias_trace	on	logs/sat_bias_trace.csv
enable_code_bias_trace	on	logs/code_bias_trace.csv (receiver DCB/IFB)
enable_sat_status_trace	on	logs/satellite_status_trace.csv
enable_dop_trace	on	logs/dop_trace.csv
enable_residual_trace	on	logs/measurement_residuals.csv
enable_wls_trace	on	logs/wls_trace.csv
enable_smoother_trace	on	logs/smoother_trace.csv
enable_mw_slip_trace	on	logs/mwwl_trace.csv
enable_if_slip_trace	on	logs/if_slip_trace.csv
enable_gf_slip_trace	on	logs/gf_slip_trace.csv
enable_turbo_edit_trace	on	logs/turbo_edit_trace.csv
enable_windup_trace	off	logs/windup_trace.csv (phase wind-up diagnostic; off by default)
kalman_filter_concise_log	off	

1.8 Cycle-slip detector registry

Table 8: Cycle-slip detectors known to `SlipDetectorRegistry`. *Active* detectors have a creator in the factory table and run when their flag is set; *staged* detectors are declared in the `SlipDetectorKind` enum but excluded from the build (the registry warns and skips them if requested). The $k\sigma$ column is the MAD gate multiplier from `CycleSlipConfig`.

Detector	Config flag	Default	$k\sigma$	Status	Combination
`MelbourneWubbena`	`estimation.cycle_slip.mw`	on	8.0	**active**	Wide-lane MW combination (primary, always available).
`Graphic`	`estimation.cycle_slip.graphic`	off	6.0	**active**	GRAPHIC (code+phase average) per-signal detector.
`ExtraWideLane`	`estimation.cycle_slip.ewl`	off	5.0	**active**	Extra-wide-lane MW for 3+ frequency satellites.
`GeometryFree`	`estimation.cycle_slip.gf`	off	–	*staged*	2nd-order time-differenced geometry-free (STAGED).
`IonoFreePhase`	`estimation.cycle_slip.ifp`	off	–	*staged*	Double-differenced ionosphere-free phase OMC (STAGED).

1.9 State-vector key schema

Table 9: Dynamic state-vector key formats, reconstructed from the `make*Key` bodies in `StateKeySchema.hpp`. Angle-bracketed fields are runtime values; literal characters are the delimiter / infix / prefix / suffix constants.

State	Key format	Builder
Ambiguity	`<satellite>_<signal>`	`makeAmbiguityKey`
NL Ambiguity	`<satellite>_NL_<band>`	`makeNlAmbiguityKey`
WL Ambiguity	`<satellite>_WL_<band_a>_<band_b>`	`makeWlAmbiguityKey`
DD Ambiguity	`DD|<cons>|<sat_i>|<code>|<pivot>`	`makeDdAmbiguityKey`
Ionosphere	`<satellite>`	`makeIonosphereKey`
IFB	`<system>_IFB<band>`	`makeIfbKey`
DCB	`<system>_DCB`	`makeDcbKey`
Glonass Per Slot IFB	`<system>_IFB<band>_k`	`makeGlonassPerSlotIfbKey`
Receiver Phase Bias	`<system>_RPB_<signal>`	`makeReceiverPhaseBiasKey`

State	Key format	Builder
Satellite Bias	`<satellite>|<signal>`	`makeSatBiasKey`
Satellite Phase Bias	`<satellite>|<signal>|PB`	`makeSatPhaseBiasKey`
Satellite Clock	`<satellite>|SC`	`makeSatClockKey`
System Time Offset	`<system>`	`makeSystemTimeOffsetKey`
Pivot Seed	`<cons>|<band>`	`makePivotSeedKey`
Vtec Sh	`SH|||<trig>`	`makeVtecShKey`